Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence

- Understanding Linear Independence in Vector Spaces
  Reading • 18 min
- Linear Independence, Basis, and Rank
  Ungraded App Item • 20 min
- Linear Independence
  Reading • 30 min

Linear Mapping

- Exploring Transformations: Understanding Linear Mappings
  Reading • 1h
- Linear Mapping: Part 1
  Video • 8 min
- Linear Mapping: Part 2
  Video • 8 min
- Linear Mapping
  Reading • 5 min
- Matrix Representation
  Reading • 6 min
- Practice Quiz: Linear Mapping
  Graded Assignment • Grade: 100%

Lesson 3: Eigenvalues and Eigenvectors

- Introduction to Eigenvalues and Eigenvectors
  Reading • 40 min
- **Eigenvectors and Eigenvalues: Examples and Applications**
  Reading • 30 min

Lesson 4: Cholesky Decomposition

- Cholesky Decomposition
  Reading • 5 min
- Solved Examples
  Reading • 10 min

Lesson 5: Singular Value Decomposition

- Introduction to Singular Value Decomposition (SVD)
  Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors
  Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Eigenvectors and Eigenvalues: Examples and Applications

0:04 / 1:25 2x

Additional Resources for Mastering Linear Algebra Concepts

To deepen your understanding of linear algebra and practice essential skills, review this selection of valuable resources. These materials include interactive lessons, visual explanations, and practice problems designed to reinforce your learning and help you apply concepts such as matrix decomposition, determinants, and eigenvalues.

1. **Khan Academy: Eigenvectors and Eigenspaces for a 3x3 Matrix**
   - This focused video tutorial explains eigenvectors and eigenspaces for a 3x3 matrix. This resource is ideal for visual learners who need clear, step-by-step explanations of these core linear algebra concepts.
2. **Algebraic and Geometric Multiplicity of Eigenvalues**
   - This resource provides a thorough breakdown of the algebraic and geometric multiplicity of eigenvalues, helping you connect these concepts to matrix properties and their broader applications in linear algebra.
3. **Harvard Lecture Notes–Linear Algebra and Vector Analysis**
   - This detailed Harvard lecture on the spectral theorem provides both theoretical explanations and applied examples to help you understand its role in linear algebra.
4. **Spectral Theorem For Dummies**
   - This user-friendly video explains the spectral theorem in simple terms, making it accessible for beginners. This is a great visual resource for anyone who prefers learning through video explanations.

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